

Mark Laptenok

A communicator and a technical leader with a fundamental background in applied mathematics and systems programming. Major SWE expertise is in communication among systems. A teacher. Interested in kernel mode programming and distributed OSes.

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EXPERIENCE

Microsoft — Senior Software Engineer

JAN 2026 - PRESENT

In the Craft & Reliability team of the Microsoft Teams desktop client for Windows and MacOS, current responsibility is to reach 99.9% reliability.

Achievements and active projects are

- designed, implemented with custom AI agents, and delivered web-content identification of failed WebView2 renderers. The statistics of most fail-prone web applications delivered to the area owner.
- researching the Out-of-Memory failures; designed a tool to track and distinguish causes of OOM states with further improvements of resilience.
- *side project*: analyzed the Teams desktop client SDLC gaps and introduced the list of solutions to the infrastructure teams. Proposed a mathematical model for greedy E2E testing of code changes (the verification is in progress)

Diebold Nixdorf — Advanced Software Engineer

JULY 2022 - DEC 2025

Achievements are

- designed endpoint's communication module for the next generation monitoring and remote control system for banking and retail; implemented the resilience part and the data serialization based on scheme exchange.
- as a DRI, refactored and delivered the communication stack of the proprietary remote control and monitoring software for banking. The project is initiated by Bank of America and took 2 years to complete.
- fixed database connection issues.

Laptenok Labs — Software Engineering Teacher and Consultant

MARCH 2019 - PRESENT

As of September 1st, 2026

- completed 4525 hours of classes in Computer Networks, Operating

SKILLS

Problem-solving

Cross-team and customer-engineer communication

Communication between computer systems

Design and performance analysis of algorithms and data structures

Communication protocol design

AWARDS

2009 - a winner of "UMNIK" (the Russian abbreviation for "Member of the Youth Research and Innovation Competition") for the research in automated natural language translation

LANGUAGES

English — C1

Czech — B2

Russian — native speaker

C, C++, Rust, Go — fluent

Systems, SQL/NoSQL Databases, Parallel Computations (pthreads, OpenMP, C++ STL, WinAPI), Algorithms and Data Structures, Software Engineering, Probability Theory, Discrete Mathematics, and different CPLs.

— tutored 172 people, including Amdocs' interns and students of Imperial College, London, The University of Bremen, The University of Auckland, Moscow State University, Moscow Engineering Physics Institute and others to improve their engineering and computer programming skills.

[Silixcon](#) — Software Engineer

MARCH 2020 - JUNE 2022

Achievements are

- designed and implemented a Python parser embedded into the C++ desktop client for proprietary motor controllers.
- designed an API, a user permission model, and a desktop client for the remote registration, update, and monitoring system for motor controllers.
- refactored the remote firmware update system for motor controllers and designed and implemented its runtime environment based on Type 3 virtualization.

PRIOR EXPERIENCE

Self-employment, Moscow, RU — Commercial Photographer, Instructional Designer

MAY 2009 - MARCH 2019

InfoTeCS Internet Trust JSC, Rostov-on-Don, RU — Network Administrator

JANUARY 2007 - JANUARY 2008

Faculty of Mathematics, Mechanics, and Computer Sciences, SfeDU, Rostov-on-Don, RU — Network Administrator

SEPTEMBER 2006 - DECEMBER 2006

EDUCATION

Faculty of Mathematics, Mechanics, and Computer Sciences, SfeDU, Rostov-on-Don, RU — 5-year Degree + 3 years of unfinished postgraduate education

SEPTEMBER 2003 - MAY 2011

The qualification is a Mathematician and Systems programmer.

Diploma and postgraduate thesis: "Pattern-based natural language translation of scientific texts". Research supervisor — [Sumbatyan, Mezhlum A.](#) Thesis opponent — [Karyakin, Mikhail I.](#)

Microsoft Training Center, Rostov-on-Don, RU — Microsoft Certified Professional

JANUARY 2007 - MARCH 2007

Credential ID 5745808 in Internet Security and Acceleration Server 2004.

PROJECTS

[E.ON](#) — Identity and Access Management System

Designed and implemented the PoC and a stack manager.

[Laptenok Labs](#) — Learning Management Platform

Designed and implemented an LMS for the distribution of self-produced learning courses in software engineering.

[Laptenok Labs](#) — Operating System for the aarch64 SoC

Implementing a distributed operating system for BCM2711 in Rust.